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Sex-specific characteristics of the trunk muscle behaviors in an asymptomatic adult cohort

Fumin Pan^{1,2†}, Jingbo Cheng^{1,2†}, Chao Kong^{1,2}, Wei Wang^{1,2} and Shibao Lu^{1,2*}

Abstract

Objectives The trunk muscles play an important role in generating motion and maintaining posture stability. We aimed to investigate the sex-dependent differences in the three-dimensional isometric and isokinetic trunk muscle strength.

Methods We included 122 asymptomatic participants (61 males and 61 females) aged from 20 to 35 years and used the Bionix Sim3 Pro device to measure the three-dimensional isometric and isokinetic trunk muscle strength.

Results Males displayed greater absolute isometric trunk torques than females ($P < 0.05$), which vanished after normalized to body weight ($P > 0.05$). The isometric torque was greatest during extension, then during flexion, lateral bending came next and axial rotation came least ($P < 0.05$). During isokinetic movements at 15°/s, males in contrast displayed lower trunk torques than females in each direction, both before and after normalized to body weight ($P < 0.05$). The isokinetic trunk torque decreased with an increasing velocity from 15°/s to 45°/s ($P < 0.05$), which showed no significant difference between 45°/s and 75°/s ($P > 0.05$). The isokinetic trunk torque was greatest during extension, then during flexion, lateral bending and then during axial rotation when moving at 15°/s ($P < 0.05$), and this sequence changed during movements at 45°/s and 75°/s.

Conclusions We created a database concerning the isometric and isokinetic trunk torques in an asymptomatic adult cohort. The results could be provided as physiological references in the future for comparison between subjects with and without trunk muscle impairments.

Keywords Trunk torque, Isometric, Isokinetic, Sex, Three-dimensional

Introduction

The trunk muscles are likely to play an important role in performing daily tasks, both in generating motion and as well in maintaining posture stability [1, 2]. When the

function of the trunk muscles is impaired, symptoms including low back pain would occur [3]. In return, low back pain could lead to a pathological reorganization of the motor cortex, making control of the trunk muscles less efficient [4, 5]. In addition, the functional impairment in the trunk muscles can lead to muscle compensation and influence athletic performance, which would potentially increase the risk of musculoskeletal injury during sports [6]. Study also demonstrated that the trunk muscle strength is associated with the sagittal spinal balance in adults, particularly the lumbar muscles [1]. Several investigators further stated that the loss of the function of the trunk muscles is associated with the

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improper posture in children [7]. Hence, it is essential to maintain the physiological function of the trunk muscles, which requires first the accurate assessment of the trunk muscle strength.

As for the assessment of the trunk muscle strength, the dynamometer is a safe, reliable, and standard instrument [8]. Due to the technical limitation, evaluation of the muscle strength was usually performed in one-single direction using dynamometry, mainly in the sagittal plane measuring trunk flexors and extensors [1–3, 9]. Nevertheless, we usually perform three-dimensional motion both in the daily life and as well in sports. However, evaluation of the trunk muscles in the three-dimensional directions is poorly addressed, which requires a special spine module. Recently, a new measuring device, the Bionix Sim3 Pro, could allow isometric, isotonic and isokinetic measurement in the axes flexion/extension, lateral flexion and rotation. During the performance of isokinetic measurements in one axis, the (compensatory) torques in the other two axes can also be measured. The device could also measure the time needed to reach maximum torque, acceleration and deceleration, average work and more [10]. Hence, we aimed to use the device to measure the isometric and isokinetic trunk muscle strength during flexion, extension, lateral bending, and axial rotation.

As reported in literature, males usually displayed greater trunk flexion and extension torques than females during isometric movements, and the differences were usually vanishing when the torques were normalized to body weight [11, 12]. However, it remains unknown whether the sex-dependent differences still exist in the coronal and axial planes and during isokinetic movements. Despite sex, the angular velocity is also a main contributor in influencing the performance of the trunk muscles. One study indicated an inverse relationship between the angular velocity and the axial rotation torque, but only males were included [13]. Furthermore, Scholars found that the trunk extension and flexion torques tended to decrease with an increasing angular velocity [6]. Yang et al. [14] demonstrated that different angular velocities showed different abilities in predicting quality of life in patients with degenerative spinal disorders. Nevertheless, it remains unknown how different velocities influence the trunk muscle strength during the three-dimensional movements. It also remains unknown whether sex displays an effect on the trunk muscle strength during lateral bending, axial rotation and during isokinetic movements. Furthermore, it is unclear whether the trunk muscle strength differs in different directions.

Hence, we first aimed to create a database of the trunk torque with asymptomatic adult subjects to be used as

reference in the future. We then aimed to investigate the sex-dependent difference of the three-dimensional isometric and isokinetic trunk muscle strength. The following hypotheses were put forward:

- (1) Males displayed greater trunk torques than females during isometric movement, which would be vanishing when normalized to body weight [11];
- (2) Males displayed similar trunk torques to females during isokinetic movement, and the sex-dependent difference would be detected after normalized to body weight [10];
- (3) During the isokinetic movement, the trunk torque tended to decrease with an increasing velocity [13];
- (4) The trunk torque was direction-dependent, with the greatest torque during movements in the sagittal plane, then in the coronal plane, and the axial plane came least.

Methods

Participants

We included subjects with the following inclusion criteria: (1) aged between 20 and 35 years; (2) asymptomatic without low back pain in the past 3 months; (3) Asian population; and (4) without obvious spinal deformity from outlook. All subjects were included through advertisements and were volunteered to participant and signed consents. From January 2024 to July 2024, we included 122 participants, consisting of 61 males and 61 females. The study was approved by the ethical committee of Xuanwu Hospital Capital Medical University (No. 2023–241-002).

Experimental device

We used the Bionix Sim3 Pro (Fig. 1) to measure the three-dimensional isometric and isokinetic trunk muscle strength. The device was verified with a high intra- and inter-reliability [10]. We also priorly tested the reliability of the device on 20 other volunteers without low back pain. Each subject was measured twice by the same examiner under the same protocol on two different days (3 days between) and we found the mean intraclass correlation coefficient between 0.88 and 0.98.

Study procedure

All subjects were upright seated during the measurements with their shoulders and pelvis fixed. The seated position was chosen, because the hip muscles were less involved, which could decrease the lumbar loads [8, 15]. Each subject was instructed to perform the three-dimensional movements, in the order of flexion, extension, left and right lateral bending, and left and

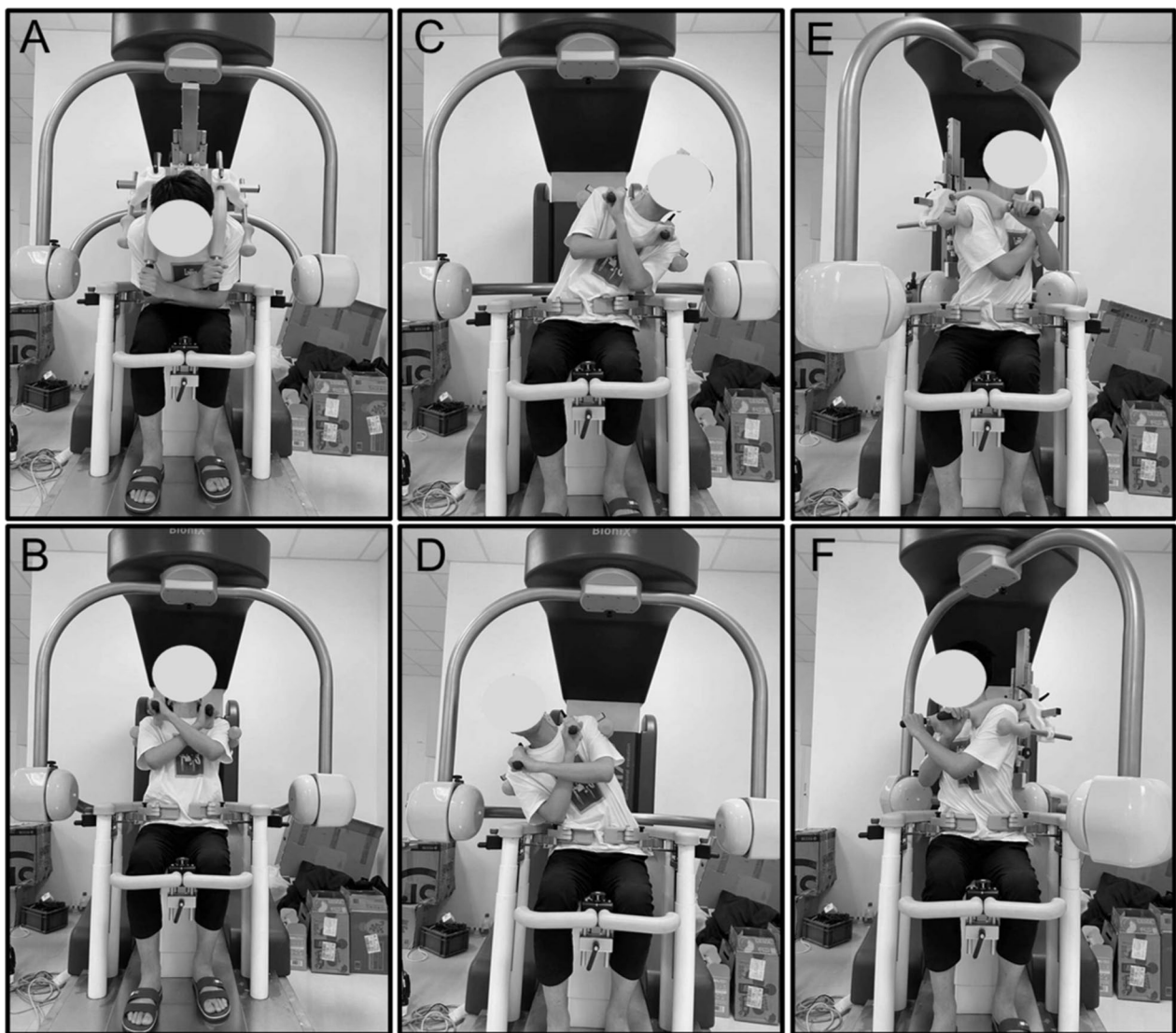


Fig. 1 Isokinetic measurement using the Bionix Sim3 Pro device during flexion **A**, extension **B**, left **C** and right **D** lateral bending, left **E** and right **F** axial rotation

right axial rotation (Fig. 1A–F). The starting position for each measurement is from a sitting, neutral position.

First, the isometric measurements were performed in the three-dimensional directions. The time length for each trial was 8 s. All subjects were motivated by verbal encouragements to reach their maximum torque. Between each two trials, subjects rested 30 s to avoid fatigue.

After the isometric measurements, a 1-min rest was given to each subject. Then the isokinetic measurements were performed randomly in three angular velocities in the three-dimensional directions: 15°/s, 45°/s and 75°/s. Between each two trials, subjects rested 30 s to avoid fatigue. Three repetitions for each measurement

were performed. The maximum value among the three repetitions was collected and expressed in newton meters (Nm) and was normalized to the respective body weight (Nm/Kg). All measurements were performed by three experienced surgeons.

Statistics

All statistical analyses were conducted using the SPSS software 19.0 (SPSS Inc., Chicago, IL). The Shapiro–Wilk and Levene tests were adopted to evaluate the normality and variances of the continuous data, respectively. When distributed normally, data were expressed as mean $\pm$ standard deviation; otherwise, median values and ranges were presented. If normally distributed, the

two-way repeated analysis of variance (ANOVA) was adopted for the isometric measurements, with sex as the inter-subject factor and direction as the intra-subject factor. The three-way repeated ANOVA was adopted for the isokinetic measurements, with sex as the inter-subject factor, and angular velocity and direction as the inter-subject factors. If not, the non-parametric Friedman test was adopted. Statistical significance was set as $P < 0.05$ and the Bonferroni's post hoc analysis was conducted for adjustments of multiple comparisons.

Results

Demographic data

One hundred and twenty-two subjects were finally included, with 61 males and 61 females (Table 1). The age of males was 24.5 ± 2.3 years, which was similar to that of 23.5 ± 1.6 years in females ($P > 0.05$). Males displayed a body weight of 73.4 ± 15.0 kg, and a body height of 175.6 ± 6.8 cm, which were both greater than those of 55.1 ± 7.6 kg and 163.4 ± 6.9 cm in females, respectively ($P < 0.05$). There existed no significant difference between males (20.7 ± 4.0 kg/m²) and females (20.6 ± 2.3 kg/m²) in BMI ($P > 0.05$).

Isometric torque

In males, the median values of the absolute isometric trunk torques were 87.3, 118.1, 71.3, 63.2, 47.7 and 49.2 Nm during flexion, extension, left and right lateral bending, left and right axial rotation, which were all greater than those of 56.9, 88.1, 49.5, 46.2, 34.1 and 35.2 Nm in females ($P < 0.05$, Fig. 2A, B), respectively. In both sexes, the absolute isometric trunk torque was greatest during extension, then during flexion, lateral bending came next and axial rotation came least ($P < 0.05$).

After normalized to body weight, the isometric trunk torques were 1.15, 1.74, 0.95, 0.91, 0.74 and 0.64 Nm/Kg during flexion, extension, left and right lateral bending, left and right axial rotation in males, which showed no significant differences compared to those of 1.05, 1.63, 1.00, 0.86, 0.66 and 0.66 Nm/Kg in females ($P > 0.05$, Fig. 2C, D), respectively. Nevertheless, the normalized isometric trunk torque was greatest during extension,

then during flexion, lateral bending came next and axial rotation came least in both sexes ($P < 0.05$).

Isokinetic torque

In the sagittal plane during flexion at 15°/s, the median value of the absolute isokinetic trunk torque was 41.8 Nm in males, which was lower than that of 49.8 Nm in females ($P < 0.05$, Fig. 3A). During extension at 15°/s, the absolute isokinetic trunk torque was 47.5 Nm in males, which was lower than that of 67.3 Nm in females ($P < 0.05$, Fig. 3B). In the coronal plane during lateral bending at 15°/s, the absolute isokinetic trunk torques were 34.9 and 30.6 Nm in males in the left and right side, which were lower than those of 46.4 and 39.3 Nm in females, respectively ($P < 0.05$, Fig. 3C, D). In the axial plane during axial rotation at 15°/s, the absolute isokinetic trunk torques were 24.5 and 26 Nm in males in the left and right side, which were, however, showed no differences to those of 23.3 and 26.9 Nm in females, respectively ($P > 0.05$, Fig. 3E, F). The absolute isokinetic trunk torque decreased in both sexes with an increasing velocity from 15°/s to 45°/s or to 75°/s ($P < 0.05$). However, there existed no difference in the absolute isokinetic trunk torque between moving at 45°/s and 75°/s ($P > 0.05$). When moving at each direction at 15°/s, the absolute isokinetic trunk torque was greatest during extension, then during flexion, lateral bending came next and axial rotation came least ($P < 0.05$); which was not the case during movements at 45°/s and 75°/s ($P > 0.05$).

After normalized to body weight, the median values of the isokinetic trunk torques at 15°/s were 0.58, 0.69, 0.47 and 0.46, 0.35 and 0.40 Nm/Kg in males, which were all lower than those of 0.93, 1.40, 0.88 and 0.78, 0.43 and 0.53 Nm/Kg in females, respectively (Fig. 4A–F, $P < 0.05$). The normalized isokinetic trunk torque decreased in both sexes with an increasing velocity from 15°/s to 45°/s or to 75°/s ($P < 0.05$). However, there existed no difference in the normalized isokinetic trunk torque between moving at 45°/s and 75°/s ($P > 0.05$). When moving at 15°/s, the normalized isokinetic trunk torque was greatest during extension, then during flexion, lateral bending came next and axial rotation came least ($P < 0.05$). However, when moving at 45°/s and 75°/s, it was greater during extension and axial rotation than during flexion and lateral bending ($P < 0.05$).

Table 1 Demographic comparisons between males and females

Item	Males	Females	<i>p</i> value
n	61	61	–
Age (years)	24.5 ± 2.3	23.5 ± 1.6	0.142
Body height (cm)	175.6 ± 6.8	163.4 ± 6.9	< 0.001
Body weight (Kg)	73.4 ± 15.0	55.1 ± 7.6	< 0.001
BMI (Kg/m ²)	20.7 ± 4.0	20.6 ± 2.3	0.767

Discussion

The trunk muscles are the active system in maintaining posture stability and as well in generating motion. When the function of trunk muscles is impaired, symptoms including low back pain or spinal imbalance would be incurred [1, 3]. Moreover, it could be used to detect deficits in specific pathologies, as well to evaluate the

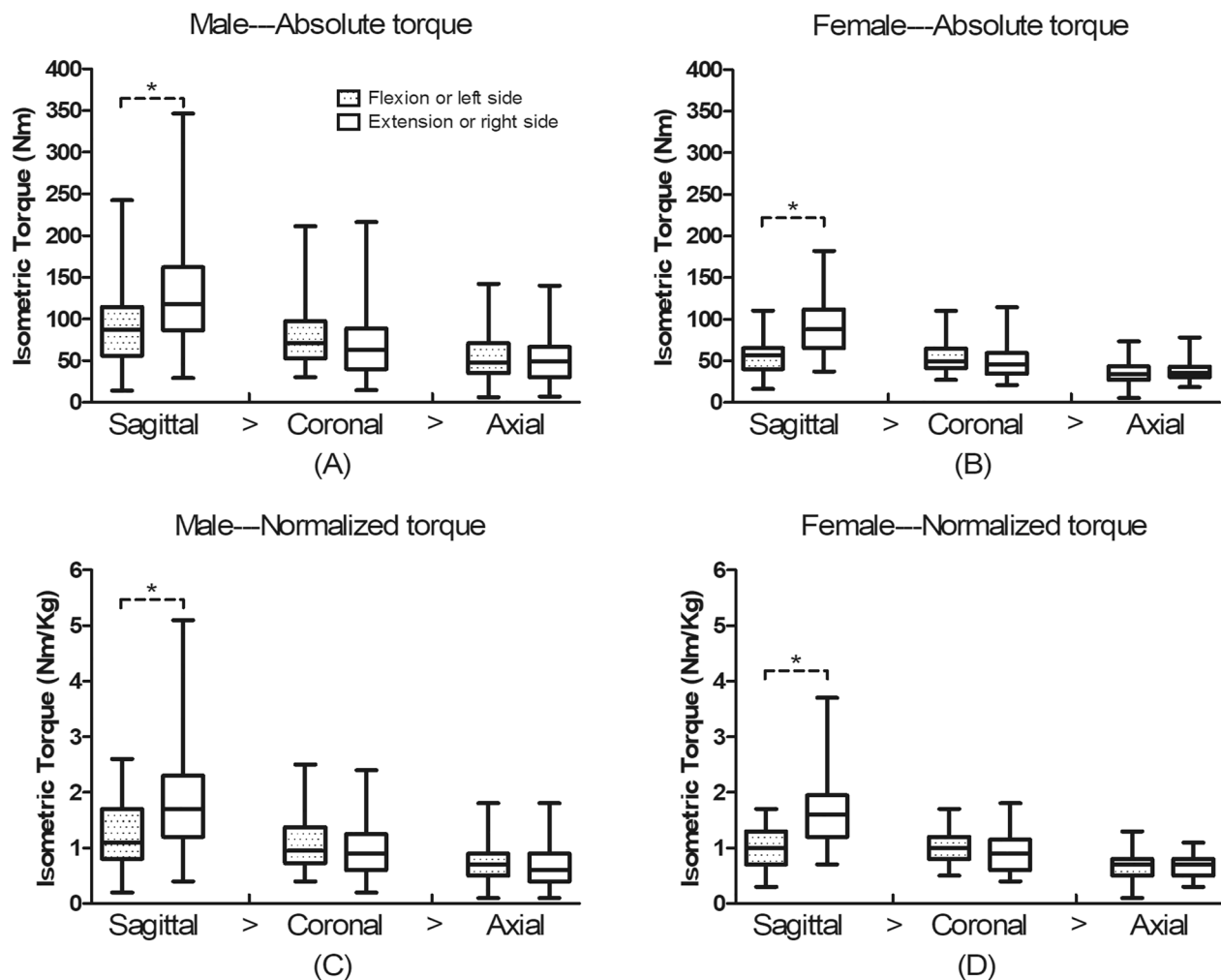


Fig. 2 Absolute **A, B** and normalized **C, D** isometric trunk torques of males and females in the sagittal, coronal and axial planes. The isometric trunk torque was greatest in the sagittal plane, then in the coronal plane and the least in the axial plane. * and > represent statistical significance at the level of 0.05

effectiveness of treatments [1, 9, 11, 16]. In athletes, the function of trunk muscles is also considered an important parameter of performance-related fitness [6, 11]. Hence, it is essential to accurately evaluate the trunk muscle strength and a normative database is needed to be used as reference. We, therefore, used the Bionix Sim3 Pro device to evaluate the isometric and isokinetic trunk muscles' torques during movements at three-dimensional planes in an asymptomatic Asian population without low back pain. We further aimed to investigate the sex- and direction-dependent differences in the trunk torque.

In the current study, the isometric trunk torques displayed great variability in both sexes, which should, therefore, be individually evaluated. The isometric torques measured in this study were lower than those measured in previous studies, as we included different populations, adopted different protocols and used

different devices [6, 13, 16]. Nevertheless, both this study and previous studies show that males displayed greater absolute isometric torque than females during movements in each direction [11, 17]. However, these differences in trunk torques were vanishing when normalized to body weight, which also agrees with previous results and accepts our first hypothesis [11, 18]. These results could be used as references in the future for comparison between subjects with and without spinal pathologies in the Asian population, for the establishment of rehabilitation strategy, and as well for the evaluation of treatment outcome.

Both in males and females, the isometric trunk torque was greatest during extension, then during flexion, lateral bending came next and it was least during axial rotation. This pattern still exists when the isometric trunk torques were normalized to body weight. We speculate that this

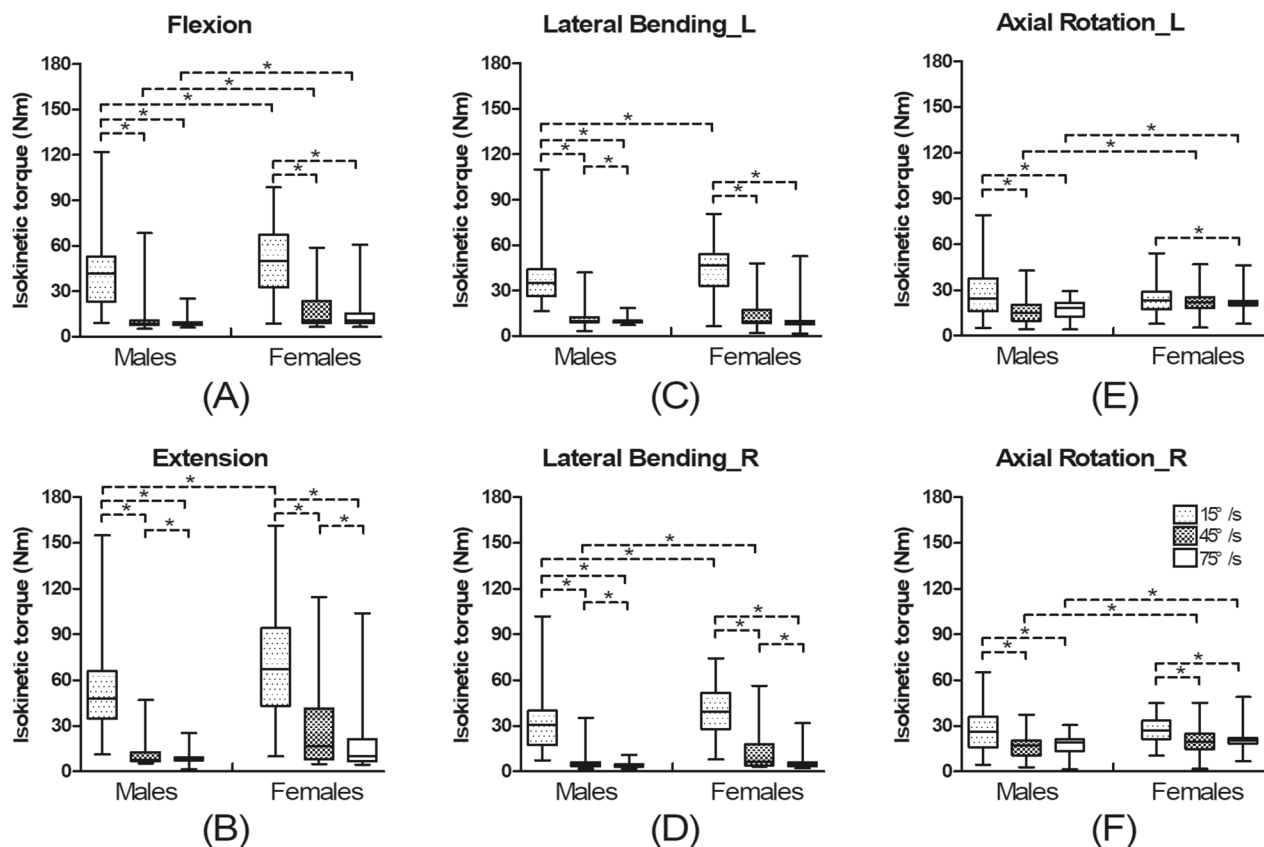


Fig. 3 Absolute isokinetic trunk torques of males and females during flexion **A**, extension **B**, left **C** and right **D** lateral bending, left **E** and right **F** axial rotation. * represents statistical significance at the level of 0.05

might be related to the anatomical characteristics of the trunk muscles and the spine. The trunk muscles are arranged longitudinally, and in addition, the orientation of the lumbar facet joints limits motion in the coronal and transverse planes. Nevertheless, the specific reasons need further investigation. We assumed the impairment of this pattern is associated with the occurrence of low back pain and the pattern of pain, as some patients demonstrated pain only during some specific movement, such as flexion or extension [19]. Multiple studies also stated that low back pain would alter the extension-flexion torque ratio [10, 11, 20], and found that a normal extension-flexion ratio of approximately 1.5 in the asymptomatic population, which is similar to the results of 1.7 in this study. Furthermore, the asymmetry of trunk motion between left and right lateral bending or axial rotation is the indication for potential deficits, which is the same case for trunk bending and rotation torques [19]. Hence, the isometric trunk torque is direction-dependent and the impairment of the physiological pattern might be associated with potential spinal pathologies.

The absolute isokinetic trunk torques during movements at 15°/s were greater in females than in

males in all directions. After normalized to body weight, the differences were even greater, which was totally in contrast to the sex-dependent difference in isometric trunk torque and rejects our second hypothesis. However, with an increasing angular velocity to 45°/s and 75°/s, the sex-dependent difference were gradually vanishing. We, therefore, assumed that among different isokinetic movement velocities, males and females adopt different strategies in activating the trunk muscle to generate motion, which causes the difference between sexes [21]. The sex-dependent difference might be related to many factors, including anatomical differences, exercise habits, etc.

During isokinetic movements, the trunk torque decreased with the angular velocity increasing from 15°/s to 45°/s in all directions, both before and after normalized to body weight, which agrees with our third hypothesis. Nevertheless, the trunk torques displayed no significant differences between isokinetic movements at 45°/s and 75°/s. Chen et al. [16] demonstrated a decreasing tendency of the trunk flexion torque and extension torque with the velocity increasing from 60°/s to 120°/s. In athletes, Ben Moussa Zouita et al. [6]

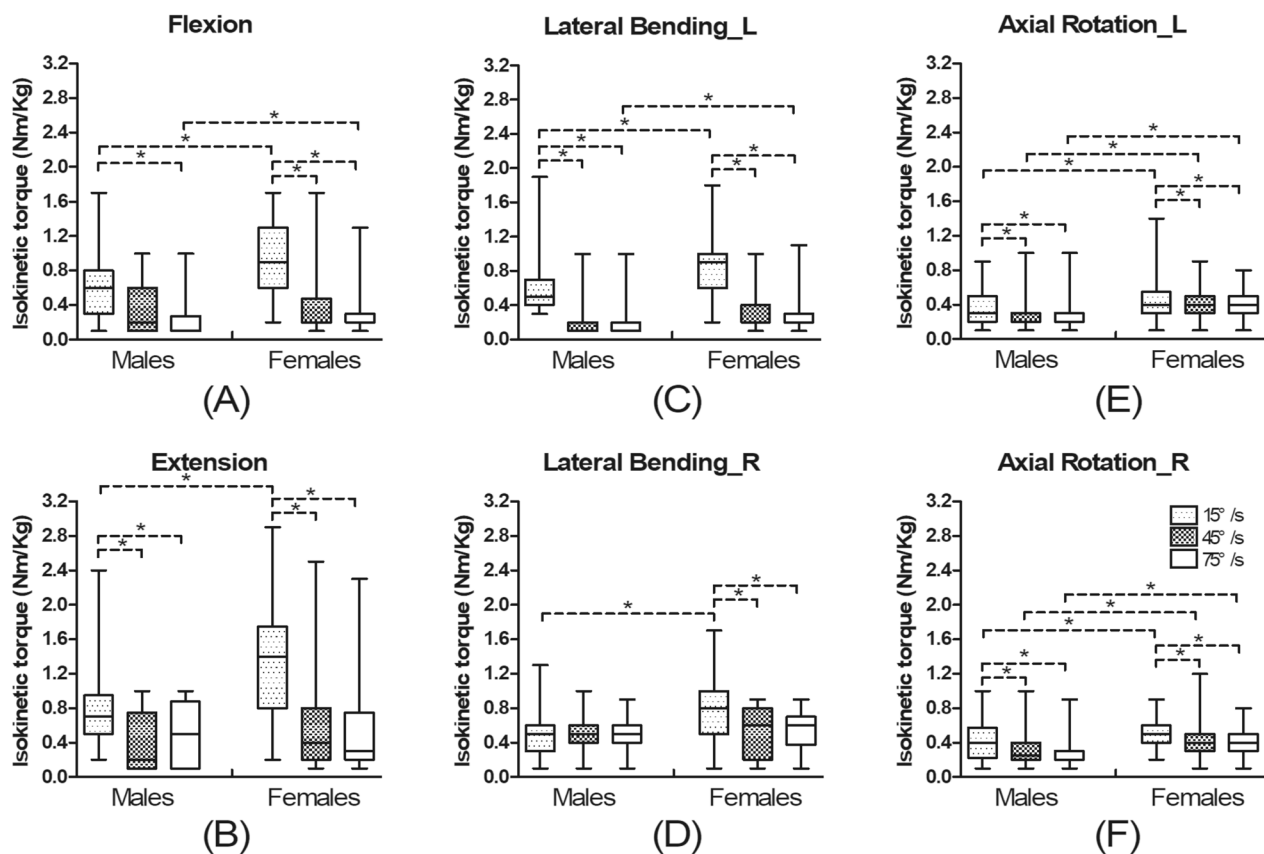


Fig. 4 Normalized isokinetic trunk torques of males and females during flexion **A**, extension **B**, left **C** and right **D** lateral bending, left **E** and right **F** axial rotation. * represents statistical significance at the level of 0.05

found that trunk extension and flexion torques tended to decrease with an increasing angular velocity. Fan et al. [13] suggested significant difference in the axial rotation torque between moving at 30°/s and 120°/s; nevertheless, no difference was detected between moving at 60°/s and at the other two velocities. However, since we have yet to standardize our measuring protocol, it is difficult to compare our results with those reported in the literature. However, we could suggest a decreasing tendency in the trunk torque in all three-dimensional direction with an increasing angular velocity, and this tendency would be stagnating when the velocity reach to a certain level.

During moving at 15°/s, the isokinetic trunk torque was greatest during extension, then during flexion, during lateral bending and then during axial rotation, which was the same as during isometric movements and accepts our last hypothesis. Nevertheless, this pattern was altered during moving at 45°/s and 75°/s and partly rejects our last hypothesis, which was also potentially altered by the different muscle synergies and should be evaluated carefully.

Before applying the result of the present study in a clinical setting, important limitations must be addressed.

We must declare that there exist several limitations of this study. First, we only included participants without low back pain and aged between 20 to 35 years. Second, only the sitting posture was investigated. Third, we only set three angular velocities and could not detect exactly at which velocity, the isokinetic movement pattern of the trunk muscle would change. Fourth but not finally, although all participants rested 30 s between each two trials, the effect of fatigue in some participants could not be ignored.

Despite the limitations, our results could be provided as physiological references in the future for comparison between subjects with and without impairments in the trunk muscles, since the impairment of these characteristics is potentially associated with the occurrence of symptoms including low back pain. Furthermore, the characteristics in the trunk muscle strength could be used for the installation of the musculoskeletal models, for the validation of the computational models, and as well for the investigation of the potential role of the trunk muscles in the occurrence of spinal pathologies, which would be beneficial for both the clinicians and researchers. Further study including

more participants and participants with low back pain or other spinal pathologies should be conducted.

In conclusion, we found that both the isometric and isokinetic trunk torques showed great variability in this study, which might be the nature characteristic of the trunk torque in the whole population. Nevertheless, there does exist the sex-, angular velocity-, and direction-dependent characteristics in the trunk torque both during isometric and isokinetic movements.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40001-025-02742-w>.

Supplementary material 1

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Not applicable

Author contributions

Fumin Pan and Shibao Lu have given substantial contributions to the conception or the design of the manuscript, Fumin Pan, Jingbo Cheng and Wei Wang to acquisition, analysis and interpretation of the data. All authors have participated to drafting the manuscript, Chao Kong revised it critically. All authors read and approved the final version of the manuscript.

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Availability of data and materials

Data is provided within the manuscript or supplementary information files.

Declarations

Ethical approval and consent to participate

The study was approved by the ethical committee of Xuanwu Hospital Capital Medical University (No. 2023-241-002). The written informed consent was obtained from all participating patients. This clinical trial was conducted in strict accordance with the Declaration of Helsinki (Edinburgh 2000 version) and relevant Chinese clinical trial research norms and regulations.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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